TSM 098

Homework 2

Square roots, order of operations, and evaluation

Topics 5, 7-8 | Estimated time: about 1 hour

Complete all problems and show your work on separate paper. Plan for about 45-50 minutes of work and 10-15 minutes to check, correct, and submit.

Use available resources to check your answers and reasoning. Complete the self-evaluation in Blackboard, upload a clear photo of your one best representative page, and retain all your written work.

Work without a calculator before checking. Give exact, simplified answers.

Roots and signs

1. Evaluate.

$$\sqrt{81}, \quad -\sqrt{81}, \quad \sqrt{0}, \quad \sqrt{\frac{25}{36}}$$

2. Evaluate and explain why the answers differ.

$$(-5)^2, \qquad -5^2$$

3. In the real-number system, does √(-16) have a value? Explain. If a problem allows complex numbers, write the principal square root using i.

4. Explain the difference between evaluating √49 and solving x squared = 49.

Order of operations

5. Evaluate. Show the order of your steps.

$$18 - 3(2 + 4)$$

6. Evaluate. Show the order of your steps.

$$-4 + 2(7 - 10)$$

7. Evaluate. Show the order of your steps.

$$24 \div (3 \cdot 2) + 5$$

8. Evaluate. Show the order of your steps.

$$3^2 - 2(-4)$$

9. Evaluate. Show the order of your steps.

$$\frac{15 - 3}{2 + 4}$$

10. Evaluate. Show the order of your steps.

$$2[5 - 3(4 - 6)]$$

Substitution and comparison

11. Evaluate at x = -3 and y = 2.

$$4x - 3y + 5$$

12. Evaluate at x = -2.

$$3x^2 + 2x - 5$$

13. Evaluate at a = -4 and b = -2.

$$\frac{a^2 - b}{3}$$

14. Evaluate both sides at x = -1, 0, and 2. Then use an algebraic property to explain whether the equation is true for every x.

$$3(x + 2) = 3x + 6$$

15. Evaluate both sides at x = -1, 0, and 2. Is the equation true for every x? Does agreement at one input prove an identity?

$$(x + 1)^2 = x^2 + 1$$

16. For x = -2, a student evaluates x squared as -4. Explain the substitution error and give the correct value.